Gas, totals and mass flow

Cold gas (stations 0-3, 13-19): γ_c = 1.4, cp_c = 1004.5 J/(kg K), R_c = 287 J/(kg K). Hot gas (4-9): γ_t = 1.3, cp_t = 1239 J/(kg K), R_t = cp_t (γ_t-1)/γ_t = 285.9 J/(kg K). R_t applies at hot stations.

$$P = \rho R T, \quad R = c_p\frac{\gamma-1}{\gamma}, \quad a = \sqrt{\gamma R T}, \quad \rho_1 v_1 A_1 = \rho_2 v_2 A_2$$

$$\dot{Q} - \dot{W} = \dot{m}\left[\left(h+\tfrac{v^2}{2}\right)_{out} - \left(h+\tfrac{v^2}{2}\right)_{in}\right]$$

W > 0 extracted (turbine), W < 0 delivered (compressor)

$$s_2 - s_1 = c_p \ln\frac{T_2}{T_1} - R\ln\frac{P_2}{P_1}, \qquad \left.\frac{T_2}{T_1}\right|_s = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}}$$

$$T_t = T + \frac{v^2}{2c_p}, \quad \frac{T_t}{T} = 1+\frac{\gamma-1}{2}M^2, \quad \frac{P_t}{P} = \left(1+\frac{\gamma-1}{2}M^2\right)^{\frac{\gamma}{\gamma-1}}$$

values in the isentropic flow tables (page 4)

$$\frac{P_{t2}}{P_{t1}} = e^{-\Delta s/R}$$

adiabatic inlet: τ_d = 1 but π_d < 1; normal shock keeps Tt, loses Pt

$$\dot{m} = \frac{P_t A\,\mathrm{MFP}(M)}{\sqrt{RT_t}}, \qquad \mathrm{MFP} = \sqrt{\gamma}M\left(1+\frac{\gamma-1}{2}M^2\right)^{-\frac{\gamma+1}{2(\gamma-1)}}$$

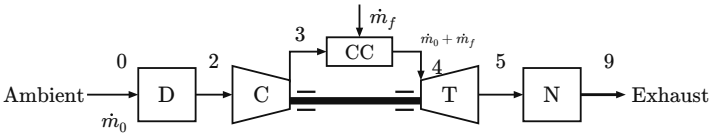
MFP max at M = 1: 0.6673 (γ 1.3), 0.6847 (γ 1.4); oblique flow: A cos(α); Pt in Pa

Station numbering (AS755)

0 freestream; 1 inlet entry; 2 inlet exit / fan face; 13 fan exit; 25 LPC exit / HPC entry; 3 HPC exit / burner entry; 4 burner exit / HPT vanes; 45 HPT exit / LPT entry; 5 LPT exit; 6, 16 mixer entry (core, fan duct); 6A mixer exit; 7 afterburner exit; 8 nozzle throat; 9 core exit; 19 fan exit.

Cycle notation and components

Turbojet block diagram



$$\tau_r = 1+\frac{\gamma_c-1}{2}M_0^2, \quad \pi_r = \tau_r^{\frac{\gamma_c}{\gamma_c-1}}, \quad \tau_\lambda = \frac{c_{pt}T_{t4}}{c_{pc}T_0}, \quad f = \frac{\dot{m}_f}{\dot{m}_3}$$

$$\pi_d = \pi_{d,\max}\eta_r, \quad \eta_r = 1\ (M_0\le 1), \quad 1-0.075(M_0-1)^{1.35}\ (1<M_0<5),$$

$$\eta_d = \frac{(P_{t2}/P_0)^{\frac{\gamma-1}{\gamma}}-1}{T_{t0}/T_0-1}$$

$$\tau_c = \pi_c^{\frac{\gamma_c-1}{\gamma_c e_c}} = 1+\frac{\pi_c^{(\gamma_c-1)/\gamma_c}-1}{\eta_c}, \qquad \dot{W}_c = \dot{m}c_{pc}(T_{t3}-T_{t2})$$

$$\varepsilon = \frac{\gamma-1}{\gamma}: \ \eta_c = \frac{\pi_c^\varepsilon-1}{\tau_c-1},\ e_c = \frac{\varepsilon\ln\pi_c}{\ln\tau_c}; \qquad \eta_t = \frac{1-\tau_t}{1-\pi_t^\varepsilon},\ e_t = \frac{\ln\tau_t}{\varepsilon\ln\pi_t}$$

e divides for a compressor (τ = π^(ε/e)), multiplies for a turbine (τ = π^(ε e))

$$f = \frac{c_{pt}T_{t4}-c_{pc}T_{t3}}{\eta_b h_{PR}-c_{pt}T_{t4}} = \frac{\tau_\lambda-\tau_r\tau_c}{\eta_b h_{PR}/(c_{pc}T_0)-\tau_\lambda}, \qquad \pi_b \approx 1-\varepsilon\frac{\gamma}{2}M_b^2$$

jet fuel h_PR ≈ 42.5-42.8 MJ/kg; stoichiometric f ≈ 0.068

$$\tau_t = 1-\frac{\tau_r(\tau_c-1)}{\eta_m(1+f)\tau_\lambda}, \qquad \pi_t = \tau_t^{\frac{\gamma_t}{(\gamma_t-1)e_t}} = \left[1-\frac{1-\tau_t}{\eta_t}\right]^{\frac{\gamma_t}{\gamma_t-1}}$$

Nozzles

$$T_{t9} = T_{t8} = T_{t5}, \quad \pi_n = \frac{P_{t9}}{P_{t5}}, \quad \eta_n = \frac{T_{t5}-T_9}{T_{t5}-T_{9s}} = \frac{1-T_9/T_{t5}}{1-(P_9/P_{t5})^{(\gamma_t-1)/\gamma_t}}$$

stations: 8 throat, 9 exit (same plane in a convergent nozzle); η_n is referred to Pt5, π_n to Pt9

$$\pi_n = \frac{P_9}{P_{t5}}\left(\frac{T_{t5}}{T_9}\right)^{\frac{\gamma_t}{\gamma_t-1}}, \qquad C_V = \frac{v_9}{v_{9s}} = \sqrt{\eta_n}$$

$$\frac{P_{t9}}{P_9} = \frac{P_0}{P_9}\pi_r\pi_d\pi_c\pi_b\pi_t\pi_n, \qquad \frac{P_{t19}}{P_{19}} = \frac{P_0}{P_{19}}\pi_r\pi_d\pi_f\pi_{fn}, \qquad \pi_{fn} = \frac{P_{t19}}{P_{t13}}$$

two spools: $\pi_c = \pi_{cL}\,\pi_{cH}$, $\pi_t = \pi_{tH}\,\pi_{tL}$; fan stream: Tt19 = Tt13, γ_c, R_c

$$M_9 = \sqrt{\frac{2}{\gamma-1}\left[\left(\frac{P_{t9}}{P_9}\right)^{\frac{\gamma-1}{\gamma}}-1\right]}, \quad T_9 = \frac{T_{t9}}{1+\frac{\gamma-1}{2}M_9^2}, \quad v_9 = M_9\sqrt{\gamma R T_9}$$

$$T_9 = T_{t5}\left[1-\eta_n\left(1-\left(\frac{P_9}{P_{t5}}\right)^{\frac{\gamma-1}{\gamma}}\right)\right],$$

$$v_9 = \sqrt{2c_p(T_{t5}-T_9)}, \quad M_9 = \sqrt{\frac{2}{\gamma-1}\left(\frac{T_{t9}}{T_9}-1\right)}$$

$$\frac{T^*}{T_t} = \frac{2}{\gamma+1}, \quad \frac{P_t}{P^*} = \left(\frac{\gamma+1}{2}\right)^{\frac{\gamma}{\gamma-1}}, \quad v^* = \sqrt{\frac{2\gamma}{\gamma+1}RT_t};$$

$$\eta_n:\ P^* = P_{t5}\left[1-\frac{1-T^*/T_t}{\eta_n}\right]^{\frac{\gamma}{\gamma-1}}$$

T*/Tt = 0.8696 (γ 1.3), 0.8333 (γ 1.4); Pt/P* = 1.832 (γ 1.3), 1.893 (γ 1.4)

$$\text{convergent: } P^* \ge P_0:\ M_9 = 1,\ P_9 = P^*,\ T_9 = T^*;$$

$$P^* < P_0:\ P_9 = P_0,\ M_9 < 1$$

isentropic: choked for Pt9/P0 ≥ 1.832 (γ 1.3), 1.893 (γ 1.4); the exit static pressure cannot fall below P*

$$\dot{m} = \frac{P_{t8}A_8\,\mathrm{MFP}(1)}{\sqrt{RT_{t8}}}, \quad A_9 = \frac{\dot{m}_9 RT_9}{P_9 v_9} = \frac{\dot{m}_9\sqrt{RT_{t9}}}{P_{t9}\,\mathrm{MFP}(M_9)}, \quad \frac{A_9}{A_8} = \frac{P_{t8}}{P_{t9}}\frac{\mathrm{MFP}(1)}{\mathrm{MFP}(M_9)}$$

isentropic divergent part: A9/A8 = A/A*(M9) (table); MFP(1) = 0.6673 (γ 1.3), 0.6847 (γ 1.4)

$$F_g = \dot{m}_9 v_9 + (P_9-P_0)A_9, \qquad dF_g = (P_9-P_0)\,dA_9 \ \Rightarrow\ F_{g,\max}\text{ at } P_9 = P_0$$

P9 > P0 under-expanded, P9 < P0 over-expanded; matched and isentropic: v9 = sqrt(2 cp Tt9 [1 - (P0/Pt9)^((γ-1)/γ)])

Convergent nozzles: subsonic civil engines and fan streams (the fan nozzle is often unchoked at cruise). C-D nozzles: high nozzle pressure ratio (supersonic flight, afterburning), usually variable area.

Thrust and performance

$$F = \dot{m}_9 v_9 - \dot{m}_0 v_0 + (P_9-P_0)A_9, \qquad A_9 = \frac{\dot{m}_9 RT_9}{P_9 v_9}, \qquad I = PA(1+\gamma M^2)$$

two streams: extra term $\dot{m}_{19}$ v19 + (P19 - P0) A19; force on a component = I_in - I_out

$$F_{inst} = F - D_{nac} - D_{add}, \qquad F_{adim} = \frac{F}{a_0\dot{m}_0}, \qquad I_{sp} = \frac{F}{\dot{m}_f g} = \frac{1}{g\,\mathrm{TSFC}}$$

$$\mathrm{TSFC} = \frac{\dot{m}_f}{F}, \quad \eta_T = \frac{\Delta E_k}{\dot{m}_f h_{PR}}, \quad \eta_P = \frac{F' v_0}{\Delta E_k}, \quad \eta_O = \eta_T\eta_P = \frac{v_0}{h_{PR}\,\mathrm{TSFC}}$$

TSFC in mg/(N s) = 1e-6 kg/(N s); uninstalled F in the efficiencies

$$\Delta E_k = \tfrac{1}{2}\left(\dot{m}_9 v_9^2 + \dot{m}_{19}v_{19}^2 - \dot{m}_0 v_0^2\right), \qquad \eta_P \approx \frac{2}{1+v_9/v_0}$$

η_P includes the pressure thrust; static (v0 = 0): $\eta_P = \eta_O = 0$

$$\frac{T_9}{T_0} = \frac{\tau_\lambda\tau_t c_{pc}/c_{pt}}{(P_{t9}/P_9)^{(\gamma_t-1)/\gamma_t}}, \qquad \frac{v_9}{a_0} = M_9\sqrt{\frac{\gamma_t R_t T_9}{\gamma_c R_c T_0}}$$

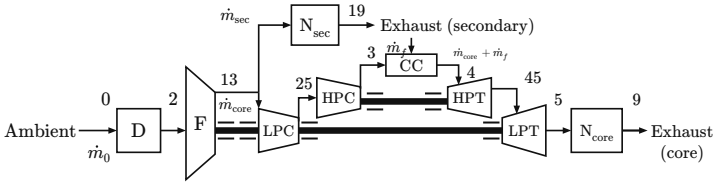
$$\frac{F}{\dot{m}_0} = a_0\left[(1+f)\frac{v_9}{a_0} - M_0 + (1+f)\frac{R_t}{R_c}\frac{T_9/T_0}{v_9/a_0}\frac{1-P_0/P_9}{\gamma_c}\right], \qquad \mathrm{TSFC} = \frac{f}{F/\dot{m}_0}$$

$$\tau_{c,\mathrm{opt}} = \frac{\sqrt{\tau_\lambda}}{\tau_r}$$

raising Tt4 and π_c raises η_T but also v9, which lowers η_P: the turbofan breaks that coupling

Turbofan and two spools

Turbofan block diagram (two spools)



$$\alpha = \frac{\dot{m}_{sec}}{\dot{m}_{core}}, \quad \dot{m}_{core} = \frac{\dot{m}_0}{1+\alpha}, \quad \dot{m}_f = f\dot{m}_{core}, \quad \mathrm{TSFC} = \frac{f}{(1+\alpha)F/\dot{m}_0}$$

$$\dot{W}_{HPC} = \eta_{m,HP}(1+f)\dot{m}_{core}c_{pt}(T_{t4}-T_{t45})$$

$$\dot{W}_{fan} + \dot{W}_{LPC} = \eta_{m,LP}(1+f)\dot{m}_{core}c_{pt}(T_{t45}-T_{t5}),$$

$$\dot{W}_{fan} = \dot{m}_0 c_{pc}(T_{t13}-T_{t2}), \quad \dot{W}_{LPC} = \dot{m}_{core}c_{pc}(T_{t25}-T_{t13})$$

$$\mathrm{OPR} = \pi_f \pi_{LPC} \pi_{HPC}, \quad T_{t,out} = T_{t,in}\left[1 + \frac{\pi^{\varepsilon} - 1}{\eta}\right],$$

$$\frac{P_{t,out}}{P_{t,in}} = \left[1 - \frac{1 - T_{t,out}/T_{t,in}}{\eta_t}\right]^{\frac{\gamma_t}{\gamma_t - 1}}$$

$$F = \dot{m}_9 v_9 + \dot{m}_{19} v_{19} - \dot{m}_0 v_0 + (P_9 - P_0)A_9 + (P_{19} - P_0)A_{19},$$

$$FR = \frac{F_{core}/\dot{m}_{core}}{F_{sec}/\dot{m}_{sec}}$$

$\dot{m}_9 = (1+f)\,\dot{m}_{core}$, $\dot{m}_{19} = \dot{m}_{sec}$

Per unit $\dot{m}_0$, core-flow quantities carry the factor $1/(1+\alpha)$.

Mixed exhaust, mixer and afterburner

Long-duct turbofan: fan duct π_{fd} = Pt16/Pt13 = 0.98-0.99, $\tau_{fd} = 1$. Kutta condition at the splitter: P6 = P16; design approximation Pt6 ≈ Pt16 closes the cycle, with α set by the pressure match.

$$\pi_t = \frac{\pi_{fd}\pi_f}{\pi_b \pi_c}, \quad \tau_t = \pi_t^{\frac{(\gamma_t - 1)e_t}{\gamma_t}}, \quad \alpha = \frac{\eta_m(1+f)\tau_\lambda(1-\tau_t) - \tau_r(\tau_c - 1)}{\tau_r(\tau_f - 1)}$$

a physical α (≥ 0) exists only for compatible π_f, π_c, Tt4; f is independent of α

$$\frac{P_{t6}}{P_0} = \pi_r \pi_d \pi_c \pi_b \pi_t, \quad \frac{T_{t6}}{T_0} = \tau_\lambda \tau_t \frac{c_{pc}}{c_{pt}}; \qquad \frac{P_{t16}}{P_0} = \pi_r \pi_d \pi_f \pi_{fd}, \quad \frac{T_{t16}}{T_0} = \tau_r \tau_f$$

$$\alpha' = \frac{\dot{m}_{16}}{\dot{m}_6} = \frac{\alpha}{1+f}, \quad c_{p6A} = \frac{c_{p6} + \alpha' c_{p16}}{1 + \alpha'},$$

$$R_{6A} = \frac{R_6 + \alpha' R_{16}}{1 + \alpha'}, \quad \gamma_{6A} = \frac{c_{p6A}}{c_{p6A} - R_{6A}}$$

cp and R are mass-averaged; γ follows from them

$$c_{p6A} T_{t6A} = \frac{c_{p6} T_{t6} + \alpha' c_{p16} T_{t16}}{1 + \alpha'},$$

$$M_{16} = \sqrt{\frac{2}{\gamma_{16} - 1}\left[\left(\frac{P_{t16}}{P_{t6}}\left(1 + \frac{\gamma_6 - 1}{2}M_6^2\right)^{\frac{\gamma_6}{\gamma_6 - 1}}\right)^{\frac{\gamma_{16} - 1}{\gamma_{16}}} - 1\right]}$$

M6 = 0.4-0.5 is a design choice

$$\frac{A_{16}}{A_6} = \alpha' \sqrt{\frac{R_{16} T_{t16}}{R_6 T_{t6}}} \frac{P_{t6}}{P_{t16}} \frac{\mathrm{MFP}(\gamma_6, M_6)}{\mathrm{MFP}(\gamma_{16}, M_{16})}, \qquad A_{6A} = A_6 + A_{16}$$

$$P_{6A} A_{6A}(1 + \gamma_{6A} M_{6A}^2) = P_6 A_6 \left[(1 + \gamma_6 M_6^2) + \frac{A_{16}}{A_6}(1 + \gamma_{16} M_{16}^2)\right]$$

implicit in M6A (subsonic root)

$$\pi_M = \frac{P_{t6A}}{P_{t6}} = \pi_{M\,\max}(1 + \alpha')\sqrt{\frac{R_{6A} T_{t6A}}{R_6 T_{t6}}} \frac{A_6}{A_{6A}} \frac{\mathrm{MFP}(\gamma_6, M_6)}{\mathrm{MFP}(\gamma_{6A}, M_{6A})}$$

$\pi_{M,max} \approx 0.97$ (friction); the rest is mixing loss

$$f_{AB} = (1 + f + \alpha)\frac{c_{pAB} T_{t7} - c_{p6A} T_{t6A}}{\eta_{AB} h_{PR} - c_{pAB} T_{t7}}, \qquad \pi_{AB} = \frac{P_{t7}}{P_{t6A}}, \quad f_{tot} = f + f_{AB}$$

fuel fractions referred to the core flow; turbojet: α = 0 and station 6 for 6A; dry: $\pi_{AB} = \tau_{AB} = 1$, f_AB = 0

$$\mathrm{MFP}(\gamma_7, M_7) = \frac{1 + \alpha + f_{tot}}{1 + \alpha + f} \mathrm{MFP}(\gamma_{6A}, M_{6A}) \sqrt{\frac{R_7 T_{t7}}{R_{6A} T_{t6A}}} \frac{P_{t6A}}{P_{t7}}$$

thermal choking if the right side > MFP(1) = 0.667 (γ 1.3); afterburner inlet M typically 0.2-0.3

$$\frac{P_{t9}}{P_9} = \frac{P_0}{P_9} \pi_r \pi_d \pi_c \pi_b \pi_t \pi_M \pi_{AB} \pi_n, \qquad \mathrm{TSFC} = \frac{f_{tot}}{(1 + \alpha)F/\dot{m}_0}$$

$$\frac{F}{\dot{m}_0} = \frac{a_0}{1 + \alpha}\left[(1 + f_{tot} + \alpha)\frac{v_9}{a_0} - (1 + \alpha)M_0 + (1 + f_{tot} + \alpha)\frac{R_{6A}}{R_c}\frac{T_9/T_0}{v_9/a_0}\frac{1 - P_0/P_9}{\gamma_c}\right]$$

lighting the afterburner ≈ doubles TSFC for ≈ 50 % more thrust (v9 grows ≈ sqrt(Tt7))

$$\frac{F_{g,mixed}}{F_{g,unmixed}} = \frac{\sqrt{(1 + \alpha)(T_h/T_c + \alpha)}}{\sqrt{T_h/T_c} + \alpha}$$

2-7 %, largest near α ≈ 2 and for hot cores; high-bypass civil engines skip it for duct weight and drag

Turboshaft

$$\tau_{tH} = 1 - \frac{\tau_r(\tau_c - 1)}{\eta_{mH}(1 + f)\tau_\lambda}, \qquad \frac{\dot{W}_{shaft}}{\dot{m}_0} = \eta_{mL}(1 + f) c_{pc} T_0 \tau_\lambda \tau_{tH}(1 - \tau_{tL})$$

$$\dot{W}_{eq} = \dot{W}_{shaft} + F v_0, \quad \mathrm{PSFC} = \frac{f}{\dot{W}_{shaft}/\dot{m}_0}, \quad \eta_T = \frac{\dot{W}_{shaft} + \Delta E_k}{\dot{m}_f h_{PR}}$$

residual jet: Pt9/P9 = (P0/P9) $\pi_r\,\pi_d\,\pi_c\,\pi_b\,\pi_{tH}\,\pi_{tL}\,\pi_n$; η_P not meaningful without the rotor

Propellers, turboprop, open rotor

$$v_p = \frac{v_0 + v_e}{2}, \quad a = \frac{v_p - v_0}{v_0}, \quad F = 2\dot{m} a v_0,$$

$$\dot{m} = \rho_0 A_D v_p, \quad \dot{W}_D = F v_p, \quad \eta_P = \frac{1}{1 + a}$$

$$v_p = \frac{v_0 + \sqrt{v_0^2 + 2F/(\rho_0 A_D)}}{2}, \quad \eta_{prop} = \eta_L \eta_P,$$

$$\eta_L = \frac{\dot{W}_D}{\dot{W}_{prop}} = \eta_{prop}(1 + a) \le 1$$

$\eta_L \le 1$ bounds η_{prop} for a given disc; v_prop = ve = 2 vp - v0

$$\eta_{prop} = 10 M_0 \eta_{\max} \ (M_0 \le 0.1), \quad \eta_{\max} \ (0.1 < M_0 \le 0.7),$$

$$\eta_{\max}\left[1 - \frac{M_0 - 0.7}{3}\right] \ (0.7 < M_0 \le 0.85)$$

$\eta_{prop,max}$ 0.80-0.90; chart $\eta_{prop} = \eta_L\,\eta_P$, no gearbox

$$F_{static} = (2\rho_0 A_D)^{1/3}(FM\,\dot{W}_{prop})^{2/3},$$

$$FM = \frac{F_{static}^{3/2}}{\sqrt{2\rho_0 A_D}\,\dot{W}_{prop}} \approx 0.70\text{-}0.80, \quad v_s = \sqrt{\frac{F}{2\rho_0 A_D}}$$

F_static ≈ W_prop^(2/3) A_D^(1/3)

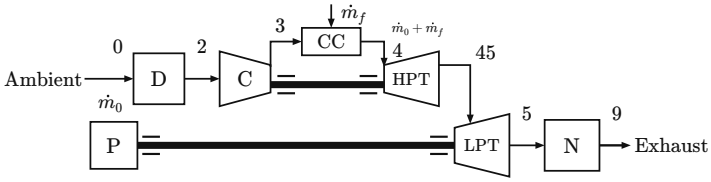
$$F\big|_{M_0=0} = \frac{10\eta_{\max}\dot{W}_{prop}}{a_0}, \qquad FM_{implied} = \left(\frac{10\eta_{\max}}{a_0}\right)^{3/2}\sqrt{\frac{\dot{W}_{prop}}{2\rho_0 A_D}}$$

FM_implied ≤ 1 within the model; measured static FM 0.70-0.80

$$\dot{W}_{prop} = \eta_g(\eta_{m,L}\dot{W}_{LPT} - \dot{W}_{LPC}),$$

$$F_{prop} = \frac{\eta_{prop}\dot{W}_{prop}}{v_0}, \quad \dot{W}_D = (1 + a)\eta_{prop}\dot{W}_{prop}$$

Turboprop block diagram



$$\beta = \frac{\dot{m}_{prop}}{\dot{m}_{jet}}, \ \dot{m}_{jet} = \frac{\dot{m}_0}{1 + \beta},$$

$$\frac{\dot{W}_{prop}}{\dot{m}_0} = \frac{\eta_g c_{pc} T_0}{1 + \beta}\left[\eta_{mL}(1 + f)\tau_\lambda \tau_{tH}(1 - \tau_{tL}) - \tau_r(\tau_{cL} - 1)\right]$$

τ_{tH} from the HP balance with $\tau_r\,\tau_{cL}$ (τ_{cH} - 1) on top

$$v_0 \frac{F}{\dot{m}_0} = \eta_{prop}\frac{\dot{W}_{prop}}{\dot{m}_0} + \frac{\gamma_c - 1}{1 + \beta} M_0 \left[(1 + f)\frac{v_9}{a_0} - M_0\right] c_{pc} T_0$$

$$\dot{m}_{prop} = \rho_0 A_D v_p, \quad \Delta E_k = \tfrac{1}{2}\dot{m}_9 v_9^2 - \tfrac{1}{2}\dot{m}_{jet} v_0^2 + \dot{W}_D,$$

$$\mathrm{TSFC} = \frac{f}{(1 + \beta)F/\dot{m}_0}, \quad \mathrm{PSFC} = \frac{f}{(1 + \beta)\dot{W}_{prop}/\dot{m}_0}$$

Mattingly uses W_prop instead of W_D: same η_O, lower η_P and higher η_T

$$J = \frac{v_0}{ND}, \quad C_P = \frac{\dot{W}_{prop}}{\rho_0 N^3 D^5}, \quad \eta_{prop} = \frac{C_F}{C_P} J, \quad M_{tip} = \frac{\sqrt{v_0^2 + (\pi N D)^2}}{a_0}$$

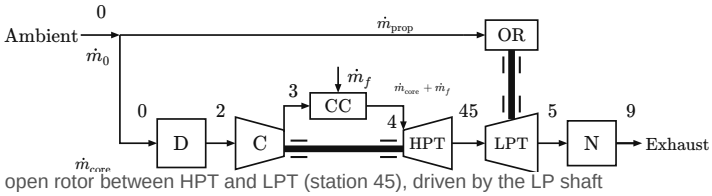
C_P = 1.76327e8 W[kW]/(σ N[rpm]^3 D^5); drag divergence near M0 0.7, swept thin blades reach ≈ 0.8

$$\tau_{tL}^* = \frac{\tau_{tH}^{-1/e_{tH}}}{\Pi}\tau_{tL}^{-\frac{1 - e_{tL}}{e_{tL}}} + A\left[1 + \frac{1 - e_{tL}}{e_{tL}}\frac{\tau_{tH}^{-1/e_{tH}}\tau_{tL}^{-1/e_{tL}}}{\Pi}\right]^2$$

$$A = \frac{(\gamma_c - 1)M_0^2/2}{\tau_\lambda \tau_{tH}(\eta_{prop}\eta_g\eta_{mL})^2}, \qquad \Pi = (\pi_r \pi_d \pi_{cL} \pi_{cH} \pi_b \pi_n)^{\frac{\gamma_t - 1}{\gamma_t}}$$

faster flight or worse propeller → larger τ_{tL}^*: more energy left in the jet

Open rotor block diagram



open rotor between HPT and LPT (station 45), driven by the LP shaft

$$\alpha_1 = \frac{\dot m_{prop}}{\dot m_{jet}}, \quad \alpha_2 = \frac{\dot m_{sec}}{\dot m_{core}},$$

$$\mathrm{TSFC} = \frac{f}{(1+\alpha_1)(1+\alpha_2)F/\dot m_0}, \quad A_D = \frac{\pi D^2}{4} - \dot m_{45}\frac{\sqrt{RT_{t45}}}{P_{t45}\mathrm{MFP}_{45}}$$

tractor propeller: A_D ≈ π D^2/4 (core streamtube neglected)

Engine sizing

$$\dot m_0 = \frac{F_{inst} + D_{nac} + D_{add}}{F/\dot m_0}, \qquad A_i = \frac{\dot m_i \sqrt{R_i T_{ti}}}{P_{ti}\,\mathrm{MFP}(\gamma_i, M_i)}$$

$\dot m_i$: $\dot m_0$ at 0-3; (1+f) $\dot m_0$/(1+α) at core 4-9; α $\dot m_0$/(1+α) at fan 13-19; (1+f+α) $\dot m_0$/(1+α) at 6A-9 (mixed). R_c, γ_c cold, R_t, γ_t hot. C-D nozzle: throat 8 and exit 9 are different areas.

Turbomachinery: Euler and conventions

$$\tau_A = \dot m(r_e v_e - r_i v_i), \qquad h_{te} - h_{ti} = \omega(r_e v_e - r_i v_i)$$

$$\text{compressor: } c_p \Delta T_t = U(v_2 - v_1),$$

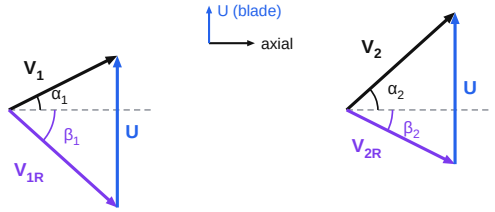
$$\text{turbine: } c_p \Delta T_t = U(v_2 + v_3), \quad U = \omega r_m$$

$$\vec V = \vec V_R + \vec U, \quad v = u\tan\alpha, \quad v_R = u\tan\beta, \quad r_m = \frac{r_t + r_h}{2}, \quad \omega = \frac{2\pi N}{60}$$

compressor: rotor 1-2, stator 2-3, v + v_R = U; turbine: nozzle 1-2, rotor 2-3, v2R = v2 - U, v3R = v3 + U (v3 positive opposite to U)

P, T, ρ are frame-invariant; Tt, Pt, M are not. Rotor: Tt,R constant (same radius). Stator: Tt constant.

Compressor stage



rotor inlet (1) **rotor exit (2) = stator inlet**

angles from the axial direction; V = V_R + U, compressor v + v_R = U; rotor 1-2, stator 2-3; repeating stage: $\alpha_1 = \beta_2$, $\beta_1 = \alpha_2$

$$c_p \Delta T_t = Uu(\tan\beta_1 - \tan\beta_2) = Uu(\tan\alpha_2 - \tan\alpha_1)$$

$$T_{t1R} = T_1 + \frac{V_{1R}^2}{2c_p} = T_{t2R}, \quad P_{t2R} = L_R P_{t1R},$$

$$P_{t3} = L_S P_{t2}, \quad L_R = 1 - \phi_{cr}\frac{\gamma P_1 M_{1R}^2}{2P_{t1R}}$$

$$R_c = \frac{T_2 - T_1}{T_3 - T_1}, \quad \Psi = \frac{c_p \Delta T_t}{U^2},$$

$$\Phi = \frac{u_1}{U} = \frac{1}{\tan\alpha_1 + \tan\beta_1}, \quad \frac{\Psi}{\Phi} = \tan\beta_1 - \frac{u_2}{u_1}\tan\beta_2$$

$$\eta_{cs} = \frac{\pi_{cs}^{\varepsilon} - 1}{\tau_{cs} - 1}, \qquad e_{cs} = \frac{\gamma - 1}{\gamma}\frac{\ln(P_{t3}/P_{t1})}{\ln(T_{t3}/T_{t1})}, \qquad M_b = \frac{U}{a_1}$$

$$D = 1 - \frac{V_e}{V_i} + \frac{|v_i - v_e|}{2\sigma V_i} \le 0.6,$$

$$D_r = 1 - \frac{V_{2R}}{V_{1R}} + \frac{|v_{1R} - v_{2R}|}{2\sigma V_{1R}}, \quad D_s = 1 - \frac{V_3}{V_2} + \frac{|v_2 - v_3|}{2\sigma V_2}$$

de Haller Ve/Vi ≥ 0.72 (rough)

$$\sigma = \frac{c}{s}, \quad s = \frac{2\pi r_m}{n_b}, \quad i = \alpha_i - \gamma_i, \quad \delta = \alpha_e - \gamma_e, \quad \delta_c = \frac{\gamma_i - \gamma_e}{4\sqrt{\sigma}}, \quad \delta_t = \frac{\gamma_i + \gamma_e}{8\sqrt{\sigma}}$$

$$\gamma_e = \frac{4\alpha_e\sqrt{\sigma} - \gamma_i}{4\sqrt{\sigma} - 1}$$

$$\text{repeating stage: } v_1 + v_2 = U, \quad c_p \Delta T_t = V_2^2 - V_1^2,$$

$$\frac{\omega r}{V_1} = \cos\alpha_1(\tan\alpha_1 + \tan\alpha_2)$$

V1R = V2, V2R = V1 (symmetric triangles); T3 = Tt3 - V1^2/(2cp)

$$\tau_{cs} = 1 + \frac{(\gamma-1)M_1^2}{1 + \frac{\gamma-1}{2}M_1^2}\left(\frac{\cos^2\alpha_1}{\cos^2\alpha_2} - 1\right),$$

$$\Psi = \frac{\tan\alpha_2 - \tan\alpha_1}{\tan\alpha_1 + \tan\alpha_2}, \quad \Phi = \frac{1}{\tan\alpha_1 + \tan\alpha_2}$$

$$D = 1 - \frac{\cos\alpha_2}{\cos\alpha_1} + \frac{\tan\alpha_2 - \tan\alpha_1}{2\sigma}\cos\alpha_2, \qquad M_3 = M_1\sqrt{\frac{T_1}{T_3}}, \qquad \pi_{cs} = \tau_{cs}^{\frac{\gamma_c}{\gamma-1}}$$

$$\pi_c = \left[\frac{T_{t1} + n_s \Delta T_t}{T_{t1}}\right]^{\frac{\gamma_c}{\gamma-1}},$$

$$n_s \approx \frac{T_{t1}\left(\pi_c^{(\gamma-1)/(\gamma e_c)} - 1\right)}{\Delta T_{t,\mathrm{stage}}}, \qquad M_{3,n} = M_1\sqrt{\frac{T_1}{T_1 + n_s \Delta T_t}}$$

stage ratios fall along the machine: $\pi_c \neq \pi_{cs}$^n; first stage set by inlet guide vanes

$$\frac{dh_t}{dr} = u\frac{du}{dr} + v\frac{dv}{dr} + \frac{v^2}{r}, \qquad \text{free vortex: } R = 1 - (1 - R_m)\left(\frac{r_m}{r}\right)^2$$

reaction lowest at the hub (turbine hub R_t typically > -0.15)

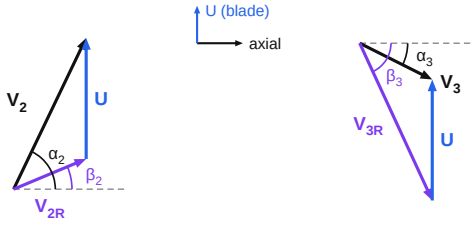
$$A = \frac{\dot m\sqrt{RT_t}}{P_t \cos\alpha\,\mathrm{MFP}(M)}, \quad r_t = \sqrt{\frac{A}{\pi\left[1 - (r_h/r_t)^2\right]}},$$

$$h = \frac{A}{2\pi r_m}, \quad \omega = \frac{U_t}{r_t}, \; N = \frac{30\omega}{\pi}$$

hub/tip 0.60-0.75 at HPC inlet, 0.90-0.92 at exit; Pt in Pa

Compressor design range	Fan / LPC	HPC
pressure ratio: 1 / 2 / 3 stages	1.5-2.0 / 2.0-3.5 / 3.5-4.5	-
inlet corrected ṁ per area, kg/(s m^2)	195-205	175-185
tip speed (m/s)	427-457	rim at exit 396-457
diffusion factor D	0.50-0.55	0.50-0.55
Φ / Ψ	-	0.45-0.55 / 0.30-0.35
first stage: hub/tip, axial M, U_tip	0.6-0.75, 0.48-0.6	350-460 m/s

Turbine stage



nozzle exit (2) = rotor inlet **rotor exit (3)**

nozzle 1-2, rotor 2-3; v3 and β_3 positive opposite to U: v2R = v2 - U, v3R = v3 + U, Ψ = (v2 + v3)/U; $\alpha_3 = 0$ gives V3 = u

$$T_{t2} = T_{t1}, \quad P_{t2} = L_S P_{t1}, \quad T_{t3R} = T_{t2R}, \quad P_{t3R} = L_R P_{t2R},$$

$$L = \frac{1}{1 + \phi_t\left[1 - (T/T_t)^{\frac{\gamma}{\gamma-1}}\right]}$$

$$\Psi = \frac{c_p \Delta T_t}{U^2} = \frac{u_2}{U}\left(\tan\alpha_2 + \frac{u_3}{u_2}\tan\alpha_3\right) = \frac{u_2}{U}\left(\tan\beta_2 + \frac{u_3}{u_2}\tan\beta_3\right),$$

$$\Phi = \frac{u_2}{U} = \frac{1}{\tan\alpha_2 - \tan\beta_2}$$

$$\text{const. } u: \; \frac{\Psi}{\Phi} = \tan\beta_2 + \tan\beta_3, \quad \frac{R_t}{\Phi} = \frac{\tan\beta_3 - \tan\beta_2}{2}, \quad R_t = \frac{T_2 - T_3}{T_{t1} - T_{t3}}$$

$$R_t = 0: \; \Psi = 2\Phi\tan\beta_2 \; (\beta_3 = \beta_2);$$

$$R_t = 0.5: \; \Psi = 2\Phi\tan\alpha_2 - 1 \; (\beta_3 = \alpha_2); \quad \alpha_3 = 0: \; \Psi = 2(1 - R_t)$$

zero swirl + 0 %: Ψ = 2, v2 = 2U, v2R = v3R = U; zero swirl + 50 %: Ψ = 1; Ψ > 2 means R_t < 0

$$\Phi = \frac{1}{\tan\alpha_2 - \tan\beta_2}, \qquad U = \sqrt{\frac{c_p \Delta T_t}{\Psi}}, \qquad u = \Phi U$$

map axes: turning $\beta_2 + \beta_3$ vs β_2; usual limits $\alpha_2 \le 70°$ and Z 0.8-1.0

$$\eta_{ts} = \frac{1 - T_{t3}/T_{t1}}{1 - (P_{t3}/P_{t1})^{(\gamma-1)/\gamma}}, \qquad e_{ts} = \frac{\gamma}{\gamma - 1}\frac{\ln(T_{t3}/T_{t1})}{\ln(P_{t3}/P_{t1})}$$

$$\alpha_3 = 0: \; \sin\alpha_2 = \frac{\Psi U}{V_2}, \quad R_t = 1 - \frac{1}{2\Psi}\left(\frac{V_2}{U}\right)^2\left[1 - \left(\frac{V_3}{V_2}\right)^2\right],$$

$$T_3 = T_2 - R_t(T_{t1} - T_{t3}), \quad M_3 = M_2\frac{V_3}{V_2}\sqrt{\frac{T_2}{T_3}}$$

$$M_2 \sin\alpha_2 = 2M_T(1-R_t),\ M_T = \frac{U}{a_2};$$

$$M_{2,opt} = \sqrt{\frac{2}{\gamma-1}\left[\left(\frac{2\left(1+\frac{\gamma-1}{2}M_1^2\right)^{\frac{\gamma+1}{\gamma-1}}}{(\gamma+1)M_1^2}\right)^{\frac{\gamma-1}{2}} - 1\right]}$$

$$Z = \frac{F_t}{F_{t\max}} = \frac{2s}{c_x}\cos^2\alpha_e\left(\tan\alpha_i + \frac{u_e}{u_i}\tan\alpha_e\right)\left(\frac{u_i}{u_e}\right)^2,$$

$$Z_r = 2\frac{s}{c_x}\cos^2\beta_3(\tan\beta_2 + \tan\beta_3)$$

design Z 0.8-1.0; equivalent loading parameter Z_r c_x/s

$$F_t = \rho u_i s(v_i + v_e) = \rho u_i^2 s\left(\tan\alpha_i + \frac{u_e}{u_i}\tan\alpha_e\right),$$

$$F_{t\max} = (P_{ti} - P_e)c_x = \frac{\rho u_e^2 c_x}{2\cos^2\alpha_e}$$

$$\sigma_c \approx \frac{\rho\omega^2 A}{4\pi}\left(1+\frac{A_t}{A_h}\right) \Leftrightarrow AN^2 = \frac{3600\times10^3}{\pi(1+A_t/A_h)}\frac{\sigma_c}{\rho}$$

N in rpm, A in m^2, σ_c/ρ in MPa/(g/cm^3) = 1000 m^2/s^2; reduced by lower N, more taper or more stages

$$\mathrm{DSF} = \frac{(\omega r_r)^2\rho}{2\sigma_d}, \qquad \sigma_{tr} = \frac{\alpha E\Delta T}{3}\left(1-\frac{r}{r_h}\right), \quad \sigma_{t\theta} = \frac{\alpha E\Delta T}{3}\left(1-\frac{2r}{r_h}\right)$$

cooled blades allow A N^2 ≈ 4-5e10 in^2 rpm^2

$$T_{t3}^{(n)} = T_{t1}^{(1)} - n_s\Delta T_t$$

only the first nozzle chokes and meters the engine flow

Choking: first nozzle M2 > 1 and rotor exit M3R < 0.9 (a 50 % first stage, Ψ = 1, cannot meter the flow). r_m = U/ω, h = A/(2 π r_m); typical LPT U 350-460 m/s.

Two-spool turbofan and propfan

$$T_{t45} = T_{t4} - \frac{c_{pc}(T_{t3} - T_{t25})}{\eta_{mH}(1+f)c_{pt}},$$

$$T_{t5} = T_{t45} - \frac{c_{pc}\left[(1+\alpha)(T_{t13} - T_{t2}) + (T_{t25} - T_{t13})\right]}{\eta_{mL}(1+f)c_{pt}}$$

Propeller on a turbofan core (propfan, open rotor): $\dot m_{jet} = \dot m_{core}$; $\tau_{LPT} = 1 - \eta_{LPT}(1 - \pi_{LPT}^{\varepsilon})$; W_LPT = (1+f) $\dot m_{core}$ cp_t Tt45 (1 - τ_{LPT}); W_prop = η_g (η_{mL} W_LPT - W_LPC); F_jet = $\dot m_9$ v9 - $\dot m_{core}$ v0 + (P9 - P0) A9.

$$\text{flight: } F_{prop} = \frac{\eta_{prop}\dot W_{prop}}{v_0},\ \dot W_D = F_{prop}v_p,\ \eta_L = \frac{\dot W_D}{\dot W_{prop}};$$

$$\text{static: } F_{prop} = (2\rho_0 A_D)^{1/3}(FM\,\dot W_{prop})^{2/3},\ v_p = \sqrt{\frac{F_{prop}}{2\rho_0 A_D}}$$

tractor propeller: A_D = π D^2/4; pusher or open rotor: annulus outside the nacelle

$$\dot m_{prop} = \rho_0 A_D v_p, \quad \text{bypass} = \frac{\dot m_{prop}}{\dot m_{core}},$$

$$\dot m_0 = \dot m_{core} + \dot m_{prop}, \quad \Delta E_k = \tfrac{1}{2}\left(\dot m_9 v_9^2 - \dot m_{core}v_0^2\right) + \dot W_D$$

Repeating compressor stage (R = 0.5)

U = ω r_m; design chart: ω r/V1 and α_2 as functions of α_1 (M1 0.4-0.6). Rotor in the relative frame, stator in the absolute frame. γ = 1.4: γ/(γ-1) = 3.5, (γ-1)/γ = 0.2857.

$$u = V_1\cos\alpha_1, \quad \tan\alpha_2 = \frac{U}{u} - \tan\alpha_1, \quad V_2 = \frac{u}{\cos\alpha_2},$$

$$V_{1R} = V_2,\ V_{2R} = V_1, \quad \Delta T_t = \frac{V_2^2 - V_1^2}{c_p}$$

$$T_{t1R} = T_1 + \frac{V_2^2}{2c_p} = T_{t2R}, \quad T_2 = T_{t1R} - \frac{V_1^2}{2c_p},$$

$$T_{t2} = T_{t1} + \Delta T_t, \quad T_3 = T_{t2} - \frac{V_1^2}{2c_p}, \quad T_{t3R} = T_3 + \frac{V_2^2}{2c_p}$$

$$P_{t1R} = P_1\left(\frac{T_{t1R}}{T_1}\right)^{3.5}, \quad P_{t2R} = P_{t1R} - \phi_{cr}\tfrac{1}{2}\rho_1 V_2^2,$$

$$P_2 = P_{t2R}\left(\frac{T_2}{T_{t1R}}\right)^{3.5}, \quad P_{t2} = P_2\left(\frac{T_{t2}}{T_2}\right)^{3.5}$$

rho1 = P1/(R T1); same as L_R = 1 - φ γ P M^2/(2 Pt)

$$P_{t3} = P_{t2} - \phi_{cs}\tfrac{1}{2}\rho_2 V_2^2, \quad P_3 = P_{t3}\left(\frac{T_3}{T_{t2}}\right)^{3.5}, \quad \pi_{cs} = \frac{P_{t3}}{P_{t1}},$$

$$\eta_{cs} = \frac{\pi_{cs}^{0.2857} - 1}{\Delta T_t/T_{t1}}, \quad e_c = \frac{0.2857\ln\pi_{cs}}{\ln(1+\Delta T_t/T_{t1})}$$

rho2 = P2/(R T2); R_c = (T2 - T1)/(T3 - T1) = 0.5 for the repeating stage

$$\Delta T_t = \frac{T_{t1}}{n_s}\left(\pi_c^{\frac{\gamma-1}{\gamma e_c}} - 1\right), \quad \pi_{cs} = \left(1+\frac{\Delta T_t}{T_{t1}}\right)^{\frac{\gamma e_c}{\gamma-1}},$$

$$\eta_c = \frac{\pi_c^{0.2857} - 1}{n_s\Delta T_t/T_{t1}}, \quad T_{t,exit} = T_{t1} + n_s\Delta T_t$$

V2^2 = V1^2 + cp dTt; n stages: π_c = (1 + n dTt/Tt1)^(γ e_c/(γ-1))

Turbine stage with constant axial velocity

$$\alpha_3 = 0,\ u \text{ const.}: \ \tan\alpha_2 = \frac{\Psi}{\Phi}, \quad \tan\beta_2 = \frac{\Psi-1}{\Phi}, \quad \tan\beta_3 = \frac{1}{\Phi}, \quad R_t = 1 - \frac{\Psi}{2}$$

50 %: Ψ = 1, $\beta_2 = 0$, $\beta_3 = \alpha_2$; 0 %: Ψ = 2, tan β_2 = tan β_3 = 1/Φ, tan α_2 = 2/Φ

$$V_2 = \frac{u}{\cos\alpha_2}, \quad V_{2R} = \frac{u}{\cos\beta_2}, \quad V_3 = u, \quad V_{3R} = \frac{u}{\cos\beta_3} = \sqrt{u^2+U^2}, \quad v_2 = \Psi U$$

$$\frac{Z_r c_x}{s} = 2\cos^2\beta_3(\tan\beta_2 + \tan\beta_3) = 2\sin 2\beta_3\ (R_t = 0), \quad = \sin 2\beta_3\ (R_t = 0.5)$$

$$T_3 = T_{t3} - \frac{u^2}{2c_p},\ P_3 = P_{t3}\left(\frac{T_3}{T_{t3}}\right)^k,\ T_{t3R} = T_3 + \frac{V_{3R}^2}{2c_p} = T_{t2R},\ P_{t3R} = P_3\left(\frac{T_{t3R}}{T_3}\right)^k = L_R P_{t2R}$$

k = γ/(γ-1) = 4.333; R = 285.9

$$T_2 = T_{t1} - \frac{V_2^2}{2c_p}, \quad P_2 = P_{t2R}\left(\frac{T_2}{T_{t2R}}\right)^k,$$

$$P_{t2} = P_2\left(\frac{T_{t1}}{T_2}\right)^k, \quad L_S = \frac{P_{t2}}{P_{t1}}, \quad M_2 = \frac{V_2}{\sqrt{\gamma R T_2}}$$

Tt2R = T2 + V2R^2/(2cp) equals Tt3R; zero reaction: T3 = T2

$$A_2 = \frac{\dot m\sqrt{RT_{t1}}}{P_{t2}\cos\alpha_2\,\mathrm{MFP}(M_2)}, \quad \frac{\sigma_c}{\rho} = \frac{\pi(1+A_t/A_h)AN^2}{3.6\times10^6}\ \left[\frac{\mathrm{MPa}}{\mathrm{g/cm^3}}\right], \quad r_m = \frac{U}{\omega}$$

Component figures of merit (Mattingly)

Component	FoM	Lvl 2	Lvl 3	Lvl 4
diffuser (A / C)	$\pi_{d,max}$	0.95 / 0.90	0.98 / 0.94	0.995 / 0.96
compressor / fan	e_c / e_f	0.84 / 0.82	0.88 / 0.86	0.90 / 0.89
burner	π_b / η_b	0.92 / 0.94	0.94 / 0.99	0.95 / 0.999
turbine uncooled / cooled	e_t	0.85 / 0.83	0.89 / 0.87	0.90 / 0.89
afterburner	π_{AB} / η_{AB}	0.92 / 0.91	0.94 / 0.96	0.95 / 0.99
nozzle (D / F)	π_n	0.97 / 0.93	0.98 / 0.95	0.995 / 0.97
shaft (alone / + PTO)	η_m	0.97 / 0.92	0.99 / 0.95	0.995 / 0.97
max Tt4 / Tt7 (K)		1390 / 1670	1780 / 2000	2000 / 2220

A = subsonic, engines in nacelles; C = supersonic, engine in airframe; D = fixed convergent, F = variable C-D nozzle. Level 3 ≈ 1985-2005 technology.

Isentropic flow tables

M	Tt/T 1.4	Pt/P 1.4	MFP 1.4	A/A* 1.4	Tt/T 1.3	Pt/P 1.3	MFP 1.3
0.10	1.0020	1.0070	0.1176	5.8218	1.0015	1.0065	0.1134
0.20	1.0080	1.0283	0.2311	2.9635	1.0060	1.0263	0.2229
0.30	1.0180	1.0644	0.3365	2.0351	1.0135	1.0598	0.3249
0.40	1.0320	1.1166	0.4306	1.5901	1.0240	1.1082	0.4164
0.50	1.0500	1.1862	0.5111	1.3398	1.0375	1.1730	0.4951
0.60	1.0720	1.2755	0.5763	1.1882	1.0540	1.2560	0.5592
0.70	1.0980	1.3871	0.6257	1.0944	1.0735	1.3598	0.6081
0.80	1.1280	1.5243	0.6595	1.0382	1.0960	1.4877	0.6419
0.90	1.1620	1.6913	0.6787	1.0089	1.1215	1.6436	0.6612
1.00	1.2000	1.8929	0.6847	1.0000	1.1500	1.8324	0.6673
1.20	1.2880	2.4250	0.6645	1.0304	1.2160	2.3337	0.6465
1.50	1.4500	3.6710	0.5822	1.1762	1.3375	3.5259	0.5610
2.00	1.8000	7.8244	0.4058	1.6875	1.6000	7.6651	0.3763

Columns for γ = 1.4 (cold) and 1.3 (hot). MFP = ṁ sqrt(R Tt)/(Pt A), dimensionless (R outside). Choked: Tt/T* = 1.2, Pt/P* = 1.893 (γ 1.4); 1.15, 1.832 (γ 1.3).

ISA standard atmosphere

T = 288.15 - 6.5 h (h in km) up to 11 km, then 216.65 K; P = 1.01325 (T/288.15)^5.2559 bar below 11 km, P = 0.22632 exp(-0.1577 (h - 11)) above; ρ = P/(R T), σ = ρ/1.225, a = sqrt(1.4 R T). At 11 km: 216.65 K, 0.2263 bar, 0.3639 kg/m^3, a = 295.1 m/s.